SSC MTS 2024 question paper with answers

15 October 2024 · Shift 2

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

A product having an MRP of ₹29,800 is on sale for ₹28,608 in-store. How much of a discount is being offered on the product?

(A) 4%
(B) 1%
(C) 2%
(D) 3%

Q.4

The mean proportional between the numbers p and q is 8. Which of the following pairs of numbers can be the values of p and q?

(A) 16 and 4
(B) 10 and 6
(C) 12 and 3
(D) 12 and 16

Q.5

Three students of class VII participated in a race during a school sports day. In a 1 km race, A wins the race with B by 100 m, and in a 300 m race, B beats C by 50 m. Then in the race of 1 km, with what margin will A beat C?

(A) 700 m
(B) 250 m
(C) 150 m
(D) 750 m

Q.6

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 72 kg and 39 kg is:

(A) 52.2
(B) 51.2
(C) 50.2
(D) 53.2

Q.8

The average weight of a certain number of apples in a basket was 0.5 kg. If 10 apples with an average weight of 0.7 kg are removed from the basket and 5 apples with an average weight of 0.6 kg are added to the basket, then the average weight of the apples in the basket decreases by 0.1 kg. What was the initial number of apples in the basket?

(A) 20
(B) 21
(C) 19
(D) 18

Q.9

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 9% per year rate of interest for 5 years.

(A) 940
(B) 860
(C) 840
(D) 900

Q.11

By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 35%?

(A) 7.55
(B) 4.55
(C) 6.55
(D) 5.55

Q.12

The average mark in an exam was 60. Due to some assessment errors, the marks of 115 students had to be changed from 85 to 50 each, and the new average is 55. The total number of students who appeared in the examination was:

(A) 763
(B) 805
(C) 902
(D) 1000

Q.13

The average weight of 30 male workers in a company is 62.5 kg and that of the remaining 20 female workers is 54.5 kg. The average weight of the total workers in the company is:

(A) 59.3 kg
(B) 58.5 kg
(C) 59.5 kg
(D) 58.3 kg

Q.14

The original breadth of a rectangular box is 25 cm. The box was then remade in such a way that its length was increased by 30% but the breadth decreased by 20% and the area increased by 100 cm^2. What is the length (in cm) of the new box?

(A) 130
(B) 91
(C) 100
(D) 109

Q.15

What is the ratio between the area of a square having side of 6 cm and that of the rhombus having diagonals of 8 cm and 6 cm?

(A) 4 : 5
(B) 3 : 2
(C) 2 : 5
(D) 3 : 4

Q.17

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 8% per year rate of interest for 6 years.

(A) 1200
(B) 1140
(C) 1240
(D) 1160

Q.19

Find the HCF of 90, 230 and 325.

(A) 2
(B) 1
(C) 5
(D) 3

Q.20

A man sold an article for ₹237, there by gaining 18.5%. The cost of the article (in ₹) was:

(A) 193
(B) 200
(C) 291
(D) 281

Reasoning Ability and Problem Solving 16 questions

Q.1

Seven people P, Q, R, S, T, U and V are sitting around a circular table, facing the centre (but not necessarily in the same order). S is sitting to the immediate left of Q. V is sitting third to the left of S. R is sitting third to the right of T. U is sitting second to the left of R. How many people are sitting between S and R when counted from the right of R?

(A) 0
(B) 1
(C) 4
(D) 2

Q.5

A, B, C, D, E and F are six friends sitting around a circular dinner table, facing the centre (but not necessarily in the same order). E is an immediate neighbour of F. B is sitting second to the left of A. D is sitting to the immediate right of F. Who is sitting third to the left of A?

(A) B
(B) C
(C) F
(D) D

Q.7

In a certain code language, 'played the piano' is written as 'lo le li' and 'sparkles a piano' is coded as 'lu li la'. How is 'piano' coded in that language?

(A) le
(B) la
(C) li
(D) lu

Q.8

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All actors are singers. All singers are dancers. Conclusions: (I) At least some singers are actors. (II) Some dancers may not be singers.

(A) Only II follows
(B) Neither I nor II follows
(C) Both I and II follow
(D) Only I follows

Q.9

What should come in place of the question mark (?) in the given series? 348 350 354 360 368 ?

(A) 374
(B) 376
(C) 378
(D) 372

Q.10

312 is related to 104 following a certain logic. Following the same logic, 363 is related to 121. To which of the following is 375 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 145
(B) 115
(C) 125
(D) 135

Q.11

What should come in place of the question mark (?) in the given series based on the English alphabetical order? SRQ ,ONM, KJI, GFE, ?

(A) ABC
(B) DCB
(C) BAZ
(D) CBA

Q.12

DJFL is related to HNJP in a certain way based on the English alphabetical order. In the same way, GMIO is related to KQMS. To which of the following is IOKQ related, following the same logic?

(A) SMOU
(B) MSOU
(C) MSUO
(D) SMUO

Q.13

In a certain code language, 'PURE' is coded as '3792' and 'BURP' is coded as '9537'. What is the code for 'B' in the given code language?

(A) 7
(B) 9
(C) 5
(D) 3

Q.14

This question has two statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which of the given conclusions logically follow(s) from the given statements. Statements: Some stones are hurting. Some stones are beautiful. Conclusions: (I) Some beautiful stones are hurting. (II) Some hurting stones are beautiful.

(A) Only conclusion (I) follows
(B) Both conclusions (I) and (II) follow
(C) Only conclusion (II) follows
(D) Neither conclusion (I) nor (II) follows

Q.15

22 is related to 132 following a certain logic. Following the same logic, 35 is related to 210. To which of the following is 18 related following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 101
(B) 110
(C) 108
(D) 113

Q.16

PWSO is related to RYUQ in a certain way based on the English alphabetical order. In the same way, MTPL is related to OVRN. To which of the following is JQMI related following the same logic?

(A) SLKO
(B) LSOK
(C) SLOK
(D) LSKO

Q.17

What will come in the place of '?' in the following equation, if '+' and '×' are interchanged and '–' and '÷' are interchanged? $18 \div 6 + 5 \times 14 - 7 = ?$

(A) 0
(B) 10
(C) –10
(D) –5

Q.18

In a certain code language, 'COKE' is coded as '1357' and 'JOKE' is coded as '5673'. What is the code for 'J' in the given code language?

(A) 6
(B) 5
(C) 7
(D) 3

Q.19

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 304 ÷ 3 + 55 × 5 – 2 = ?

(A) 904
(B) 901
(C) 902
(D) 903

Q.20

In a certain code language, 'write this sentence' is coded as 'aw pm nl' and 'this is mine' is coded as 'nl wb cd'. How is 'this' coded in that code language?

(A) nl
(B) aw
(C) wb
(D) cd

General Awareness 25 questions

Q.1

Who received the National Nritya Shiromani award 2019 for promoting Kathak dance form across the world?

(A) Aparna Satheesan
(B) Anita Sharma
(C) Anindita Neogy Anaam
(D) Jatin Goswami

Q.2

Which Indian dynasty is associated with the creation of highly skillful bronze sculptures?

(A) Pallava
(B) Mughal
(C) Mauryan
(D) Chola

Q.3

Which Ministry of India launched the Pradhan Mantri Rojgar Protsahan Yojana (PMRPY)?

(A) Ministry of Finance
(B) Ministry of Labour & Employment
(C) Ministry of Social Justice and Empowerment
(D) Ministry of Rural Development

Q.4

Which famous medieval singer was born in Gwalior, Madhya Pradesh in the year 1506?

(A) Swami Haridas
(B) Tansen
(C) Gopal Nayek
(D) Chaitanya Deva

Q.5

Which of the following wars took place in the year 1761?

(A) Wandiwash
(B) Panipat II
(C) Panipat III
(D) Buxar

Q.6

In June 2023, who was appointed as the new Research and Analysis Wing (RAW) chief?

(A) Samant Goel
(B) Arun Kant
(C) Ravi Sinha
(D) Rajiv Kumar

Q.7

Which great Indian classical dancer choreographed the songs for Madhuri Dixit in 'Devdas' and 'Dil To Pagal Hai' movies?

(A) Birju Maharaj
(B) Farah Khan
(C) Prabhu Deva
(D) Ganesh Hegde

Q.8

Who was the successor of Vima Kadphises in Kushana dynasty?

(A) Kujula Kadphises
(B) Kanishka
(C) Kharavela
(D) Menander

Q.9

In which year was the first-ever motion to remove a Supreme Court Justice signed by the members of Parliament?

(A) 1993
(B) 1995
(C) 1988
(D) 1991

Q.10

The Sarda River rises in the _____ glacier in the Nepal Himalayas.

(A) Milam
(B) Bokhar Chu
(C) Machoi
(D) Siachen

Q.11

Minor points called 'Yuko' are part of which of the following sports?

(A) Judo
(B) Skating
(C) Lawn bowls
(D) Boxing

Q.12

In 1917, Mahatma Gandhi started his first satyagraha in India against the forced cultivation of indigo by Europeans in which of the following districts of British India?

(A) Champaran
(B) Buxar
(C) Ahmedabad
(D) Kheda

Q.13

International Day of Non-violence is observed on _____.

(A) 2 October
(B) 26 November
(C) 26 January
(D) 15 August

Q.14

What is Gross Domestic Product (GDP) in the context of National Income Accounting?

(A) The total income of a country's government
(B) The total exports of a country
(C) The total savings of a nation
(D) The total value of all goods and services produced within a country's borders in a specific period

Q.15

What is the product of the following chemical reaction? $NaCl + H_2O + CO_2 + NH_3$

(A) Bleaching powder
(B) Gypsum
(C) Baking soda
(D) Washing soda

Q.16

For which English novel was Ruskin Bond honoured with the Sahitya Akademi Award in 1992?

(A) A Little Book of Happiness
(B) The Blue Umbrella
(C) The Room on the Roof
(D) Our Trees Still Grow in Dehra

Q.17
Which of the following communicable diseases is caused by bacteria?

(A) Cholera
(B) Influenza
(C) Dengue
(D) Malaria

Q.18
What is the frequency (f) of a wave with a period of 0.04 seconds?

(A) 40 Hz
(B) 25 Hz
(C) 50 Hz
(D) 100 Hz

Q.19
After the _____ of Hakim Mirza, Kabul was annexed by Akbar and given to Raja Man Singh as 'jagir'.

(A) marriage
(B) victory
(C) death
(D) capture

Q.20
Which of the following iron and steel plants is located in Odisha?

(A) Bokaro
(B) Vijay Nagar
(C) Vishakhapatnam
(D) Rourkela

Q.21
At which sports event in India is the Raja Bhalindra Singh Trophy awarded?

(A) SGFI Games
(B) Senior National Kabaddi Championship
(C) National Games of India
(D) Khelo India Youth Games

Q.22
Suspended solids in water can increase the haziness of the water. Which of the following represents the measurement of haziness of water?

(A) Viscosity
(B) Salinity
(C) Solubility
(D) Turbidity

Q.23
In which region of Bangladesh do the Chakma people predominantly live, among the country's various indigenous ethnic groups?

(A) Rajshahi Division
(B) Sylhet Division
(C) Sundarbans Mangrove Forest
(D) Chittagong Hill Tracts

Q.24
In January 2024, Prime Minister Narendra Modi paid tributes to Savitribai Phule on her birth anniversary. In which field did Savitribai Phule make significant contributions, as highlighted by Prime Minister Modi?

(A) Literature and poetry
(B) Education and social reform
(C) Politics and government leadership
(D) Military strategy and warfare

Q.25
If a citizen is helping the government in protecting the monuments in his/her area, which of the following fundamental duties is he/she performing?

(A) Cherishing and following noble ideas
(B) Providing opportunities to obtain education
(C) Promoting scientific temper
(D) Valuing and preserving the rich heritage and culture

English Language and Comprehension 25 questions

Q.1

Select the most appropriate option to fill in the blank. I will try to _____ because he is a good friend of mine.

(A) get him off the hook
(B) stop him in his tracks
(C) show him the door
(D) avoid him like the plague

Q.2

Select the most appropriate ANTONYM of the underlined word in the given sentence. He was not able to answer as the question was “ambiguous”.

(A) Unsuitable
(B) Deep
(C) Clear
(D) Superficial

Q.3

The bat flew silently through the night, searching for insects to eat. The bat in this context refers to _____.

(A) a measurement unit in digital information systems
(B) the act of delivering a forceful strike or hit
(C) a wooden tool utilised in sports like baseball or cricket
(D) a mammal known for nocturnal flight and insect consumption

Q.4

The following sentence has been split into four segments. Identify the segment that contains a spelling error. This item is in the public domain / and can be used, / copied and modefied / without any restrictions.

(A) and can be used,
(B) copied and modefied
(C) without any restrictions
(D) This item is in the public domain

Q.5

Select the most appropriate synonym of the word given in brackets to fill in the blank. My teacher's _____ (stern) warning made all the students quiet and calm.

(A) pliant
(B) biased
(C) watchful
(D) strict

Q.6

Select the most appropriate option to fill in the blank. The teacher asked us to _____ our homework by tomorrow.

(A) complete
(B) completed
(C) completes
(D) completing

Q.7

Identify the word with a spelling error in the following sentence. Select the option that correctly rectifies the error. She had swam the English Channel twice.

(A) swimmed
(B) swum
(C) swimming
(D) swim

Q.8

Select the most appropriate synonym of the underlined word. Samira is a lady who needs “extrinsic” motivation and will not work on her own.

(A) Adventurous
(B) Intriguing
(C) Amicable
(D) Outward

Q.9

Select the most appropriate present tense form of the verb to fill in the blank. They _____ football every day.

(A) had been playing
(B) will play
(C) plays
(D) play

Q.10

Select the option that can be used as a one-word substitute for the given group of words. The allowance that is given by a husband to his wife on separation

(A) Dowry
(B) Affianced
(C) Matrimonial
(D) Alimony

Q.11

Select the most appropriate option to fill in the blank. The library books were long _____, their pages bearing the weight of the passage of time.

(A) overdate
(B) overdrew
(C) overdue
(D) overdo

Q.12

Select the most appropriate option to fill in the blank. I remember_____ when she used to wash dishes in a restaurant.

(A) the area
(B) the business
(C) the profile
(D) the time

Q.13

Select the INCORRECTLY spelt word. She recieved a beautiful surprise from her partner.

(A) Received
(B) Surprise
(C) Partner
(D) Beautiful

Q.14

Select the most appropriate option that can substitute the underlined words in the given sentence. Thousands of people were “killed by” the train accident that took place last Wednesday.

(A) killed in
(B) killed to
(C) killed from
(D) killed of

Q.15

Select the most appropriate option that can substitute the underlined segment in the given sentence. Dolphins are named among the “intelligent” animals on earth.

(A) most intelligent
(B) more intelligent
(C) intelligentest
(D) intelligenter

Q.16

Select the most appropriate meaning of the given idiom. Caught in a time warp

(A) To remain unchanged from a time in the past
(B) To make extra effort
(C) To be extremely happy
(D) To avoid talking about what’s important

Q.17

Select the most appropriate option that can substitute the underlined words in the given sentence. In the beauty contest, Parul won the award for being the “beautiful” one.

(A) less beautiful
(B) most beautiful
(C) least beautiful
(D) more beautiful

Q.18

Select the most appropriate option with the correct usage of the preposition to substitute the underlined segment in the given sentence. World Press Freedom Day will be dedicated to the importance of journalism and “freedom in expression” in the context of the current global environmental crisis.

(A) freedom of expression
(B) freedom at expression
(C) freedom from expression
(D) freedom on expression

Q.19

Change the following sentence into positive degree. Mercury is the smallest planet in the solar system.

(A) The most smallest planet in the solar system is mercury.
(B) Mercury is smaller than any other planet in the solar system.
(C) No other planet is as small as Mercury in the Solar System.
(D) Mercury is the smallest planet than any other one.

Q.20

Select the sentence that has a grammatical error.

(A) She saw an excellent episode of the series.
(B) After a tiring day, I had a cup of coffee with Sudhir.
(C) There is a man sitting beside cat.
(D) She asked an easy question, and I took an hour to crack it.

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Forests provide essential ecosystem services (1)_____ benefit both the environment and human society. One crucial service is carbon sequestration, as forests absorb and store large amounts of carbon dioxide, helping mitigate climate change. Additionally, forests regulate water cycles (2)_____ influencing precipitation patterns and maintaining water quality. (3)_____, forests offer recreational opportunities, cultural significance, and economic benefits through timber production and (4)_____. Thus, preserving and sustainably managing forests is imperative for maintaining these (5)_____ ecosystem services.

Select the most appropriate option to fill in blank number 1.

(A) that
(B) this
(C) than
(D) then

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Forests provide essential ecosystem services (1)_____ benefit both the environment and human society. One crucial service is carbon sequestration, as forests absorb and store large amounts of carbon dioxide, helping mitigate climate change. Additionally, forests regulate water cycles (2)_____ influencing precipitation patterns and maintaining water quality. (3)_____, forests offer recreational opportunities, cultural significance, and economic benefits through timber production and (4)_____. Thus, preserving and sustainably managing forests is imperative for maintaining these (5)_____ ecosystem services.

Select the most appropriate option to fill in blank number 2.

(A) off
(B) between
(C) about
(D) by

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Forests provide essential ecosystem services (1)_____ benefit both the environment and human society. One crucial service is carbon sequestration, as forests absorb and store large amounts of carbon dioxide, helping mitigate climate change. Additionally, forests regulate water cycles (2)_____ influencing precipitation patterns and maintaining water quality. (3)_____, forests offer recreational opportunities, cultural significance, and economic benefits through timber production and (4)_____. Thus, preserving and sustainably managing forests is imperative for maintaining these (5)_____ ecosystem services.

Select the most appropriate option to fill in blank number 3.

(A) Moreover
(B) As long as
(C) Though
(D) Whatever

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Forests provide essential ecosystem services (1)_____ benefit both the environment and human society. One crucial service is carbon sequestration, as forests absorb and store large amounts of carbon dioxide, helping mitigate climate change. Additionally, forests regulate water cycles (2)_____ influencing precipitation patterns and maintaining water quality. (3)_____, forests offer recreational opportunities, cultural significance, and economic benefits through timber production and (4)_____. Thus, preserving and sustainably managing forests is imperative for maintaining these (5)_____ ecosystem services.

Select the most appropriate option to fill in blank number 4.

(A) industrialisation
(B) ecotourism
(C) degradation
(D) deforestation

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Forests provide essential ecosystem services (1)_____ benefit both the environment and human society. One crucial service is carbon sequestration, as forests absorb and store large amounts of carbon dioxide, helping mitigate climate change. Additionally, forests regulate water cycles (2)_____ influencing precipitation patterns and maintaining water quality. (3)_____, forests offer recreational opportunities, cultural significance, and economic benefits through timber production and (4)_____. Thus, preserving and sustainably managing forests is imperative for maintaining these (5)_____ ecosystem services.

Select the most appropriate option to fill in blank number 5.

(A) vitally
(B) vitalise
(C) vitality
(D) vital

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(B)**	Q.8 **(A)**	Q.9 **(D)**
Q.11 **(B)**	Q.12 **(B)**	Q.13 **(A)**	Q.14 **(A)**	Q.15 **(B)**	Q.17 **(A)**
Q.19 **(C)**	Q.20 **(B)**				

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.5 **(D)**	Q.7 **(C)**	Q.8 **(C)**	Q.9 **(C)**	Q.10 **(C)**
Q.11 **(D)**	Q.12 **(B)**	Q.13 **(C)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(B)**
Q.17 **(C)**	Q.18 **(A)**	Q.19 **(D)**	Q.20 **(A)**		

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(C)**
Q.7 **(A)**	Q.8 **(B)**	Q.9 **(D)**	Q.10 **(A)**	Q.11 **(A)**	Q.12 **(A)**
Q.13 **(A)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(D)**	Q.17 **(A)**	Q.18 **(B)**
Q.19 **(C)**	Q.20 **(D)**	Q.21 **(C)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(B)**
Q.25 **(D)**					

English Language and Comprehension

Q.1 **(A)**	Q.2 **(C)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(D)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(B)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(A)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 4%

Discount = ₹29,800 − ₹28,608 = ₹1,192, and 1,192/29,800 × 100 = 4%.

Q.4 · Answer (A) 16 and 4

The mean proportional of p and q is √(pq), so pq = 8^2 = 64. Of the pairs, only 16 × 4 = 64.

Q.5 · Answer (B) 250 m

When A runs 1,000 m, B runs 900 m. B and C run in the ratio 300 : 250, so while B runs 900 m, C runs 900 × 250/300 = 750 m. A beats C by 1,000 − 750 = 250 m.

Q.6 · Answer (B) 51.2

(40 + 49 + 56 + 72 + 39)/5 = 256/5 = 51.2 kg.

Q.8 · Answer (A) 20

With n apples the total is 0.5n kg. Removing 10 × 0.7 = 7 kg and adding 5 × 0.6 = 3 kg leaves 0.5n − 4 kg in n − 5 apples at 0.4 kg each: 0.5n − 4 = 0.4n − 2, so n = 20.

Q.9 · Answer (D) 900

SI = 2,000 × 9 × 5/100 = ₹900.

Q.11 · Answer (B) 4.55

The cost price is 20/1.10 = ₹18.18. A 35% profit needs a selling price of 20 × 1.35/1.10 = ₹24.545, so the price must rise by ₹4.545, about ₹4.55.

Q.12 · Answer (B) 805

Each of the 115 changes lowers the total by 35, a fall of 115 × 35 = 4,025. The average falls by 5, so there were 4,025/5 = 805 students.

Q.13 · Answer (A) 59.3 kg

(30 × 62.5 + 20 × 54.5)/50 = (1,875 + 1,090)/50 = 2,965/50 = 59.3 kg.

Q.14 · Answer (A) 130

With length L, the area goes from 25L to 1.3L × 20 = 26L, a rise of L cm^2. So L = 100 and the new length is 1.3 × 100 = 130 cm.

Q.15 · Answer (B) 3 : 2

Square: 6^2 = 36 cm^2. Rhombus: ½ × 8 × 6 = 24 cm^2. 36 : 24 = 3 : 2.

Q.17 · Answer (A) 1200

SI = 2,500 × 8 × 6/100 = ₹1,200.

Q.19 · Answer (C) 5

90 = 2 × 3^2 × 5, 230 = 2 × 5 × 23 and 325 = 5^2 × 13; the only common factor is 5.

Q.20 · Answer (B) 200

The selling price is 118.5% of the cost, so the cost = 237/1.185 = ₹200.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) 1

Seated clockwise (each person's left, facing the centre): Q, S, P, R, V, U, T, which meets every condition. Going from R to its right (anticlockwise), only P sits before S: one person.

Q.5 · Answer (D) D

Seated clockwise (each person's left, facing the centre): A, C, B, D, F, E, with B second to A's left, D to F's immediate right and E beside F. In a circle of six, third to the left of A is the seat opposite A: D.

Q.7 · Answer (C) li
'piano' is the only word in both phrases and 'li' the only code in both, so piano = li.

Q.8 · Answer (C) Both I and II follow
All actors are singers, so some singers are actors (I). Nothing says every dancer is a singer, so it remains possible that some dancers are not singers (II). Both follow.

Q.9 · Answer (C) 378
The differences are 2, 4, 6, 8, so the next is 10: 368 + 10 = 378.

Q.10 · Answer (C) 125
Each number is divided by 3: 312 ÷ 3 = 104 and 363 ÷ 3 = 121. Likewise 375 ÷ 3 = 125.

Q.11 · Answer (D) CBA
Each group is three consecutive letters in reverse, and each group starts 4 letters before the last one: S, O, K, G, C. The next group is CBA.

Q.12 · Answer (B) MSOU
Each letter moves 4 places forward: DJFL → HNJP and GMIO → KQMS. So IOKQ → MSOU.

Q.13 · Answer (C) 5
P, U and R are in both words, as are 3, 7 and 9 in both codes. The letter left in BURP is B and the digit left in its code is 5, so B = 5.

Q.14 · Answer (D) Neither conclusion (I) nor (II) follows
The stones that are hurting and the stones that are beautiful may be different stones, so neither conclusion follows.

Q.15 · Answer (C) 108
Each number is multiplied by 6: 22 × 6 = 132 and 35 × 6 = 210. Likewise 18 × 6 = 108.

Q.16 · Answer (B) LSOK
Each letter moves 2 places forward: PWSO → RYUQ and MTPL → OVRN. So JQMI → LSOK.

Q.17 · Answer (C) –10
After interchanging the signs: 18 – 6 × 5 + 14 ÷ 7 = 18 – 30 + 2 = –10.

Q.18 · Answer (A) 6
O, K and E are in both words, as are 3, 5 and 7 in both codes. The letter left in JOKE is J and the digit left in its code is 6, so J = 6.

Q.19 · Answer (D) 903
After interchanging the signs: 304 × 3 – 55 ÷ 5 + 2 = 912 – 11 + 2 = 903.

Q.20 · Answer (A) nl
'this' is the only word in both sentences and 'nl' the only code in both, so this = nl.

General Awareness

Q.1 · Answer (C) Anindita Neogy Anaam
The US-based Kathak dancer Anindita Neogy Anaam received the National Nritya Shiromani award at the 10th Cuttack Mahotsav in Odisha in January 2019, for preserving and promoting Kathak across the world.

Q.2 · Answer (D) Chola
The Cholas are famed for their bronzes, cast by the lost-wax method; the dancing Shiva, Nataraja, is the best known.

Q.3 · Answer (B) Ministry of Labour & Employment
The Ministry of Labour and Employment launched PMRPY in 2016; through the EPFO, the government paid part of employers' provident-fund contributions for new employees to encourage hiring.

Q.4 · Answer (B) Tansen
Tansen, the singer of Akbar's court and one of its 'nine gems', was born near Gwalior; 1506 is one of the years given for his birth.

Q.5 · Answer (C) Panipat III
The Third Battle of Panipat, between the Marathas and Ahmad Shah Abdali, was fought on 14 January 1761. Wandiwash was fought in 1760, the Second Battle of Panipat in 1556 and Buxar in 1764.

Q.6 · Answer (C) Ravi Sinha
Ravi Sinha, a 1988-batch IPS officer, was appointed chief of RAW in June 2023, succeeding Samant Goel.

Q.7 · Answer (A) Birju Maharaj
The Kathak maestro Pandit Birju Maharaj choreographed Madhuri Dixit's dances in 'Dil To Pagal Hai' (1997) and 'Devdas' (2002), including 'Kaahe Chhed Mohe'.

Q.8 · Answer (B) Kanishka
Vima Kadphises was succeeded by Kanishka, the greatest Kushana ruler. Kujula Kadphises founded the dynasty; Kharavela ruled Kalinga and Menander was an Indo-Greek king.

Q.9 · Answer (D) 1991
The motion to remove Justice V. Ramaswami, the first against a Supreme Court judge, was signed by 108 Lok Sabha members in 1991; when it was put to the vote in 1993 it failed.

Q.10 · Answer (A) Milam
NCERT's 'India: Physical Environment', which this question follows, says the Sarda (Saryu) rises in the Milam glacier in the Nepal Himalayas, where it is called the Goriganga, and is known as the Kali along the India-Nepal border. (The Milam glacier itself lies in Uttarakhand's Pithoragarh district.)

Q.11 · Answer (A) Judo
A 'yuko' was a minor score in judo, below 'waza-ari' and 'ippon'.

Q.12 · Answer (A) Champaran
Gandhi's Champaran Satyagraha (1917), in Bihar, took up the cause of peasants forced by European planters to grow indigo under the tinkathia system.

Q.13 · Answer (A) 2 October
Since 2007 the UN has marked 2 October, Mahatma Gandhi's birthday, as the International Day of Non-Violence.

Q.14 · Answer (D) The total value of all goods and services produced within a country's borders in a specific period
GDP is the market value of all final goods and services produced within a country's borders in a given period, usually a year.

Q.15 · Answer (C) Baking soda
In the Solvay process sodium chloride, water, carbon dioxide and ammonia react to give sodium hydrogencarbonate ($NaHCO_3$), baking soda, and ammonium chloride.

Q.16 · Answer (D) Our Trees Still Grow in Dehra
Ruskin Bond received the Sahitya Akademi Award in 1992 for 'Our Trees Still Grow in Dehra'. 'The Room on the Roof', his first novel, won the John Llewellyn Rhys Prize.

Q.17 · Answer (A) Cholera
Cholera is caused by the bacterium Vibrio cholerae. Influenza and dengue are caused by viruses, and malaria by a protozoan, Plasmodium.

Q.18 · Answer (B) 25 Hz
Frequency is the reciprocal of the period: f = 1/T = 1/0.04 = 25 Hz.

Q.19 · Answer (C) death
Akbar's half-brother Mirza Muhammad Hakim ruled Kabul; Akbar annexed it after Hakim's death in 1585.

Q.20 · Answer (D) Rourkela
Rourkela Steel Plant, set up with German help, is in Odisha. Bokaro is in Jharkhand, Vijayanagar in Karnataka and Visakhapatnam in Andhra Pradesh.

Q.21 · Answer (C) National Games of India
The Raja Bhalindra Singh Trophy goes to the state or team that tops the medal table at the National Games.

Q.22 · Answer (D) Turbidity
Turbidity is the cloudiness of water caused by suspended particles; salinity is its salt content.

Q.23 · Answer (D) Chittagong Hill Tracts
The Chakmas, mostly Buddhist, live mainly in the Chittagong Hill Tracts in south-eastern Bangladesh.

Q.24 · Answer (B) Education and social reform
Savitribai Phule, born on 3 January 1831, opened schools for girls in Pune with Jyotirao Phule from 1848 and worked against caste and gender discrimination.

Q.25 · Answer (D) Valuing and preserving the rich heritage and culture
Article 51A(f) asks every citizen to value and preserve the rich heritage of the country's composite culture; protecting monuments serves that duty.

English Language and Comprehension

Q.1 · Answer (A) get him off the hook
To 'get someone off the hook' is to free them from trouble or blame, which one would try to do for a good friend.

Q.2 · Answer (C) Clear
'Ambiguous' means open to more than one meaning; its opposite is 'clear'.

Q.3 · Answer (D) a mammal known for nocturnal flight and insect consumption
'Flew silently through the night, searching for insects' points to the animal: the bat is a flying mammal that hunts insects at night.

Q.4 · Answer (B) copied and modefied
The correct spelling is 'modified'.

Q.5 · Answer (D) strict
'Stern' means serious and strict: 'strict'.

Q.6 · Answer (A) complete
'Asked us to' is followed by the base form of the verb: 'to complete our homework'.

Q.7 · Answer (B) swum
'Had' takes the past participle, and the forms are 'swim, swam, swum': 'She had swum the English Channel twice.'

Q.8 · Answer (D) Outward
'Extrinsic' means coming from outside, as opposed to 'intrinsic': 'outward'.

Q.9 · Answer (D) play
A habit ('every day') takes the simple present, and the plural 'they' takes 'play'.

Q.10 · Answer (D) Alimony
Alimony is the allowance paid to a spouse after separation or divorce.

Q.11 · Answer (C) overdue
Library books kept past their return date are 'overdue'; 'long overdue' fits pages that bear the weight of time.

Q.12 · Answer (D) the time
'I remember the time when ...' recalls a period in the past; 'when' points to a time.

Q.13 · Answer (A) Received
'Recieved' is misspelt: the word is 'received' ('i' before 'e' except after 'c').

Q.14 · Answer (A) killed in
People are killed 'in' an accident, not 'by' it.

Q.15 · Answer (A) most intelligent
'Among' points to a group, so the superlative is needed: 'among the most intelligent animals on earth'.

Q.16 · Answer (A) To remain unchanged from a time in the past
Someone or something 'caught in a time warp' has not changed from an earlier time, as if time had stood still.

Q.17 · Answer (B) most beautiful
Winning the award picks out one of all the contestants, so the superlative: 'for being the most beautiful one'.

Q.18 · Answer (A) freedom of expression
The fixed phrase is 'freedom of expression', as in 'freedom of speech'.

Q.19 · Answer (C) No other planet is as small as Mercury in the Solar System.
A superlative becomes positive with 'No other ... as ... as': 'No other planet is as small as Mercury.' 'Smaller than any other' is the comparative degree.

Q.20 · Answer (C) There is a man sitting beside cat.
A singular countable noun needs an article: 'sitting beside a cat' (or 'the cat').

Q.21 · Answer (A) that
The relative pronoun 'that' introduces the clause describing the services: 'services that benefit both the environment and human society'.

Q.22 · Answer (D) by
Forests regulate water cycles 'by' influencing precipitation: 'by' with an -ing form gives the means.

Q.23 · Answer (A) Moreover
'Moreover' adds a further point to those already made, following 'Additionally' in the sentence before.

Q.24 · Answer (B) ecotourism
Economic benefits come 'through timber production and ecotourism'; degradation and deforestation are harms, not benefits.

Q.25 · Answer (D) vital
An adjective is needed before 'ecosystem services': 'vital', meaning essential.

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